Midnight Buildathon / Wave 2

# TeaPass

A private proof that a tea lot was issued, blended, and tested.

The recipe stays off the ledger.

Local devnet evidence, 19 October 2026.

The record that has to stay private

# Lots are blended.

A finished batch is made from other batches.

The factory will not publish suppliers, weights, or the formula.

A buyer still needs to check that the recorded weights obeyed the rule.

The proof checks that digital record.

It does not prove where the physical leaf grew.

What the chain is allowed to see

# Hide the recipe.

Private: quantities, salts, owner secrets, the lab reading, the blend split.

Public: commitments, nullifiers, issuer and lab authority,

the batch reference, the material class, and whether the claim still holds.

Supply-graph anonymity is not claimed.

Public note metadata does reveal the batch reference and material.

Settled on the local Midnight devnet

# 100 + 80 = 40 + 40 + 100

Authorized issuer. Two made-tea notes: 100 kg and 80 kg.
The blend consumes both and creates two 40 kg remainders
and one 100 kg blend. Mass must balance.
Blend transaction 005dc40a settled in block 391.
A second spend fails against indexed state, before submission.
Contract 774a4f1b on the local undeployed network.

A claim, not a food-safety badge

# The reading stays hidden.

An approved lab anchors a salted result to the batch commitment.

The circuit proves that reading is under the stated demo threshold.

Judges see the policy, the lab, and the status. Not the measurement.

The replay then revokes the lab.

The saved public claim becomes INVALID.

How a judge reproduces it

# Compile it yourself.

scripts/replay-judges.sh

Pins Compact 0.31.1, starts the local node, indexer, and proof server, and replays issuance, the blend, the lab proof, and revocation.

Receipts: artifacts/midnight/inventory-local.json

Walkthrough: docs/JUDGES.md and docs/PROOF.md.

Said plainly

# What this is not.

Not Midnight preview, preprod, or mainnet.

Not a browser that submits these transactions yet.

A finalized job does not, by itself, prove that declared lineage edges match the private witnesses.

Negative cases fail in the circuit, before the node accepts them.

This wave

# New Compact rules.

Inventory issuance, transfer, split, blend, and processing.

A batch-bound laboratory threshold proof, with revocation.

An API that reconciles finality and will not fake a verified flag.

Apache-2.0. The judge compiles from this repository.